Group II and Group III cross sections: check against `Calculation_main.tex`

Contents

1 Answer 1

2 Reference 2

3 M1: the sign of Group III, Rows III, V, VI 3

4 M2: the plus prescription 4

5 M3: Group II, Row IV 4

6 M4: index placement for a general adjoint U 5

7 M5: integration measures 5

8 M6: points in `Calculation_main.tex` 6

9 Checks 6

1 Answer

How it was checked. Every displayed equation of `Group_II.tex` and `Group_III.tex` was transcribed into code and compared with a first-principles evaluation of the same quantity.

- **Reference.** The reference is built from the coefficients of `Calculation_main.tex`: $\mathbb{A}^{(1)}$, $\mathbb{B}_2$ (one-ρ and two-ρ parts), $\mathbb{C}_1$. They enter the two-gluon amplitude $\mathcal{A}_{ij}^{ab}$ and the combination of Eq. (CS2).
- **Evaluation.** The comparison is made at random transverse points, with random SU(3) adjoint Wilson lines and random charges.
- **Integrated lines.** Lines after a p^+, w' or z integration are compared with the exact integral of the (already verified) line before it, over $\Lambda < p^+ < \vee - k^+$.
- **No limits.** No limit is taken in $\vee$, k^+, p^+ or Λ. The only conditions used are $\vee > p^+, k^+ > \Lambda$ and the support of the coefficients.

The checks are listed in Sec. 9. Equation numbers such as II(1.12) or III(1.7) are those of your compiled `Group_II.tex` and `Group_III.tex`. The corrected notes are `Group_II_corrected.tex` and `Group_III_corrected.tex` (changes in blue, removed text in red).

Correct as written (with your convention $U^{bc} = U^{cb}$):

- **Group II, Row I.**
  - Part 1: the explicit form and the p^+ integration (both blocks).
  - Part 2: the explicit form, the Wilson-line reduction, the expansion of the bracket, and all six integrated parts 2a–2f.

  All logarithms there are exact.
- **Group II, Row IV** up to and including the reduction of the Wilson lines: the explicit form, the w and w' integration with its Jacobian $(\frac{p^+ + k^+}{k^+})^4$, the polarization sum, and the phase.
- **Group III, Rows I and II**: the explicit forms and the p^+ integration.

- **Group III, Rows III, V, VI**, apart from the overall sign (M1):
  - the explicit forms;
  - the w' integration (Jacobian, $\mathcal{K}(w'-z) \to \frac{k^+}{p^++k^+}\mathcal{K}(y'-z)$, phase $e^{-ik\cdot(y'-w)}e^{-ik\cdot(y'-z)p^+/k^+}$);
  - the coefficients of the $\log(\vee/\Lambda)$ terms of the plus prescription;
  - the Wilson-line simplification of Row V.

Mistakes.

M1 **Group III, Rows III, V and VI have the wrong overall sign**, in every line including the $\log(\vee/\Lambda)$ terms (Sec. 3).

- In Eq. (CS2) these rows enter as $-4\,\mathfrak{C}'^{\dagger}\mathfrak{B}$, $-2\,\mathfrak{C}'^{\dagger}\mathbb{X}'_1$ and $-2\,\mathfrak{C}'^{\dagger}\mathbb{X}'_2$. Your set-up lines have +4, +2, +2.
- The fix F5 found earlier from the JIMWLK matching (Row III, part 2) is the same sign error.
- Your three-ρ list (`Evolution_three_rho.tex`) already carries the correct sign for the $\log(\vee/\Lambda)$ term of Row V.

M2 **The plus prescription changes the lower limit of the p^+ integral from Λ to** 0 (Rows III parts 1 and 2, V, VI). This drops $\int_0^\Lambda dp^+(\dots)$, which is not small unless $\Lambda \ll k^+$ and $\Lambda|\mathbf{k}||y'-z| \ll k^+$. In the tests it changes the result by 0.3–10%. With the lower limit kept at Λ the plus prescription is exact (Sec. 4).

M3 **Group II, Row IV, the z integration and the fragmentation form** (Sec. 5):

(a) The constant in the logarithm is wrong. In the scheme of Group I Row II the z integral gives $-\pi[\frac{2}{\epsilon} + 2\gamma_E + \log\frac{(y'-y)^2\mu^2}{4}]$. In your notation, with $1/\bar\epsilon$ for the infrared pole, that is $[\frac{1}{\bar\epsilon} - \log\frac{(y'-y)^2\mu^2 e^{2\gamma_E}}{4}]$, not $[\frac{1}{\bar\epsilon} - \log(4\pi^2\mu^2(y'-y)^2)]$.

(b) The limits of the ξ integral are wrong:
- $\int_1^{k^+/\vee}$ uses $\xi = 1$ ($p^+ = 0$) instead of $\xi = \frac{k^+}{k^++\Lambda}$ ($p^+ = \Lambda$);
- the endpoint logarithm $\log\frac{\vee}{\Lambda} + \log\frac{k^+}{\vee}$ misses $\log(1-\frac{k^+}{\vee}) + \log(1+\frac{\Lambda}{k^+})$.

(c) The fragmentation form is wrong in three places:
- the measure must be $d\xi/\xi^3$ (not $d\xi/\xi^2$) when $d^3N_{\rm LO}$ is evaluated at $(k^+/\xi, \mathbf{k}/\xi)$;
- the $\delta(1-\xi)$ term of $D_{g/g}$ adds the LO cross section, which is not part of this row;
- the $\delta(1-\xi)$ part of P_{gg} is not produced by this row.

M4 **Index placement of the adjoint Wilson lines** (Sec. 6). Many brackets are correct only if $U^{bc} = U^{cb}$, as in your convention. A real adjoint Wilson line is orthogonal, $U^\dagger = U^T$, but not symmetric: $U^T = U$ together with $U^TU = 1$ would mean $U^2 = 1$. With a general U the equations hold only after the placements listed in Sec. 6. With those, every line was checked for a general U.

M5 **Integration measures** (typos): variables missing after a δ-function integration, or integrated variables still listed (Sec. 7).

M6 `Calculation_main.tex` (Sec. 8):

- the second term of $\mathbb{X}_1$ needs $U^{db}(v)$, not $U^{\dagger db}(v)$;
- the Θ-function of $\mathbb{B}_2$ should read $p^+ < \vee - k^+$;
- two smaller index slips.

2 Reference

From `Calculation_main.tex`, with $\mathbb{A} \equiv \mathbb{A}_2$:

- $\mathbb{A}^a_i(P,-W) = \frac{ig}{\sqrt{2}\pi\sqrt{P}}\int_X \mathcal{K}^i(X-W)\rho^a(X)$; $\bar{\mathbb{A}}$ has $\rho \to U\rho$.
- $\mathbb{B}^{bc}_{2ji}(k^+,-w;p^+,-z)$: the gluon (c,i,k^+) is at w and the gluon (b,j,p^+) at z. It has a one-ρ part ($f^{cba}\rho^a(x)$) and a two-ρ part ($\{\rho^c(y),\rho^b(x)\}$).
- $\mathbb{C}^{dac}_{1jik}(P, y-w; p^+, w-z)$, with $P = p^+ + k^+$, contains $\delta^{(2)}(y - w + \frac{p^+}{P}(w-z))$.

The two-gluon amplitude is $\frac{2}{\sqrt{2\pi}}\int_{w'} e^{-ikw'}\{\mathfrak{B}+\frac{1}{2}\mathbb{X}'_1+\frac{1}{2}\mathbb{X}'_2+\mathfrak{C}\}$ with

$$\mathfrak{B}^{ab}_{ij} = U^{ac}(w)\mathbb{B}^{bc}_{2ji} - U^{\dagger bc}(z)\bar{\mathbb{B}}^{ca}_{2ji}, \tag{2.1}$$

$$\mathbb{X}'^{\,ab}_{1ij} = U^{\dagger bc}(z)\bar{\mathbb{A}}^c_j(p^+,-z)\bar{\mathbb{A}}^a_i(k^+,-w) - U^{ac}(w)\bar{\mathbb{A}}^d_j(p^+,-z)\,U^{db}(z)\,\mathbb{A}^c_i(k^+,-w), \tag{2.2}$$

$$\mathbb{X}'^{\,ab}_{2ij} = U^{\dagger bc}(z)\bar{\mathbb{A}}^a_i(k^+,-w)\bar{\mathbb{A}}^c_j(p^+,-z) - \bar{\mathbb{A}}^a_i(k^+,-w)\mathbb{A}^b_j(p^+,-z), \tag{2.3}$$

$$\mathfrak{C}^{ab}_{ij} = -\mathfrak{C}'^{ab}_{ij}, \qquad \mathfrak{C}'^{ab}_{ij} = \int_y \Big[\bar{\mathbb{C}}^{dac}_{1jik}U^{\dagger bd}(z)U^{ce}(y)\mathbb{A}^e_k(P,-y) - U^{\dagger bd}(z)\bar{\mathbb{C}}^{dac}_{1jik}\bar{\mathbb{A}}^c_k(P,-y)\Big]. \tag{2.4}$$

So Eq. (CS2) reads

$$\frac{d^3N^{(\mathrm{II})}}{d^3k} = \frac{4}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\Big\{\mathfrak{B}^\dagger\mathfrak{B}+\tfrac{1}{4}\mathbb{X}'^\dagger_1\mathbb{X}'_1+\tfrac{1}{4}\mathbb{X}'^\dagger_2\mathbb{X}'_2+\mathfrak{C}'^\dagger\mathfrak{C}'\Big\}, \tag{2.5}$$

$$\frac{d^3N^{(\mathrm{III})}}{d^3k} = \frac{4}{(2\pi)^3}\int\frac{dp^+}{2\pi}\int_{z,w,w'} e^{-ik(w'-w)}\Big\{\tfrac{1}{2}\mathbb{X}'^\dagger_1\mathfrak{B}+\tfrac{1}{2}\mathbb{X}'^\dagger_2\mathfrak{B}-\mathfrak{C}'^\dagger\mathfrak{B}+\tfrac{1}{4}\mathbb{X}'^\dagger_2\mathbb{X}'_1-\tfrac{1}{2}\mathfrak{C}'^\dagger\mathbb{X}'_1-\tfrac{1}{2}\mathfrak{C}'^\dagger\mathbb{X}'_2\Big\}+\text{c.c.} \tag{2.6}$$

The p^+ range of every row checked here is $\Lambda < p^+ < \vee - k^+$. For $\mathfrak{B}$ this is because the pair comes from the window; for $\mathfrak{C}$ the parent has $p^+ + k^+ < \vee$.

3 M1: the sign of Group III, Rows III, V, VI

Where the sign comes from. In $\mathcal{A}^{ab}_{ij}$ the splitting after the target enters as $+\int_y[\bar{\mathbb{C}}U^\dagger U\mathbb{A}^\dagger_1 - U^\dagger\bar{\mathbb{C}}\bar{\mathbb{A}}^\dagger_1]$, from $\bar{\mathbb{C}}^\dagger_2 = -\bar{\mathbb{C}}_1$ and from the $\mathbb{C}_1\mathbb{A}$ term of $\mathbb{B}^\dagger_1$. With $\mathbb{A}^\dagger_1(P,y) = -\mathbb{A}_2(P,-y)$ this is $-\mathfrak{C}'$. The minus sign is the same as the one in LO, where the emission after the target is $\bar{\mathbb{A}}^\dagger_1 = -\bar{\mathbb{A}}_2$. Hence:

- Group III, Row III: $-\frac{4}{(2\pi)^3}\int\frac{dp^+}{2\pi}\ldots[\mathfrak{C}'^\dagger][\mathfrak{B}]$;
- Rows V and VI: $-\frac{2}{(2\pi)^3}\int\frac{dp^+}{2\pi}\ldots[\mathfrak{C}'^\dagger][\mathbb{X}'_{1,2}]$.

Your set-up lines have $+$, and every following line inherits it. Numerically, with $U^{bc} = U^{cb}$, every line of these rows is exactly -1 times the first-principles value (Sec. 9); all other rows give $+1$.

The corrected $\log(\vee/\Lambda)$ terms (with the exact lower limit, M2). They are your terms with the sign reversed:

$$\begin{aligned}
\text{III, part 1:}\quad & -\frac{1}{(2\pi)^3}\frac{ig^4f^{c'd'a}}{16\pi^5}\frac{1}{k^+}\Big[\log\frac{\vee}{\Lambda}+\log\Big(1-\frac{k^+}{\vee}\Big)\Big]\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2}\\
&\times\big[U^{e'c'}(y')U^{d'b}(z)\rho^{e'}(x') - U^{bd'}(z)U^{c'e'}(x')\rho^{e'}(x')\big]\\
&\times\big[U^{ac}(w)\{\rho^c(y),\rho^b(x)\} - U^{bc}(z)U^{ad}(y)U^{ce}(x)\{\rho^d(y),\rho^e(x)\}\big],
\end{aligned} \tag{3.1}$$

$$\begin{aligned}
\text{III, part 2:}\quad & -\frac{1}{(2\pi)^3}\frac{g^4}{8\pi^5}\frac{1}{k^+}\Big[\log\frac{\vee}{\Lambda}+\log\Big(1-\frac{k^+}{\vee}\Big)\Big]\int_{w,x',y',z,y,x} e^{-ik\cdot(y'-w)}\Big[\frac{(x-w)\cdot(x'-y')\,(z-w)\cdot(y'-z)}{(z-w)^2(x-w)^2(y'-z)^2(x'-y')^2}\\
&+\frac{(x-z)\cdot(y'-z)\,(x-w)\cdot(x'-y')}{2(x-z)^2(x-w)^2(y'-z)^2(x'-y')^2}\Big]\big[f^{c'd'a}U^{e'c'}(y')U^{d'b}(z)\rho^{e'}(x') - f^{c'd'a}U^{bd'}(z)U^{c'e'}(x')\rho^{e'}(x')\big]\\
&\times\big[f^{cbd}U^{ac}(w)\rho^d(x) - f^{acd}U^{bc}(z)U^{de}(x)\rho^e(x)\big],
\end{aligned} \tag{3.2}$$

$$\begin{aligned}
\text{V:}\quad & -\frac{1}{(2\pi)^3}\frac{ig^4f^{c'ca}}{8\pi^5}\frac{1}{k^+}\Big[\log\frac{\vee}{\Lambda}+\log\Big(1-\frac{k^+}{\vee}\Big)\Big]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2}\\
&\times\big[U^{e'c'}(y')\rho^{e'}(x') - U^{c'e'}(x')\rho^{e'}(x')\big]\big[U^{ce}(x)U^{ad}(y)\rho^e(x)\rho^d(y) - U^{ad}(w)U^{ce}(x)\rho^e(x)\rho^d(y)\big],
\end{aligned} \tag{3.3}$$

$$\begin{aligned}
\text{VI:}\quad & -\frac{1}{(2\pi)^3}\frac{ig^4f^{c'd'a}}{8\pi^5}\frac{1}{k^+}\Big[\log\frac{\vee}{\Lambda}+\log\Big(1-\frac{k^+}{\vee}\Big)\Big]\int_{w,y',z,y,x,x'} e^{-ik\cdot(y'-w)}\frac{(y'-z)\cdot(x-z)\,(x'-y')\cdot(y-w)}{(y'-z)^2(x'-y')^2(y-w)^2(x-z)^2}\\
&\times\big[U^{e'c'}(y')U^{d'b}(z)\rho^{e'}(x') - U^{bd'}(z)U^{c'e'}(x')\rho^{e'}(x')\big]\\
&\times\big[U^{bc}(z)U^{ad}(y)U^{ce}(x)\rho^d(y)\rho^e(x) - U^{ad}(y)\rho^d(y)\rho^b(x)\big].
\end{aligned} \tag{3.4}$$

Each comes with c.c. (the c.c. reverses the order of the charges). The full corrected rows are in `Group_III_corrected.tex`.

4 M2: the plus prescription

For the $\frac{1}{p^+}$ terms, write $g(p^+)$ for everything that multiplies $\frac{1}{p^+}$. In Row III part 2 this includes $\frac{1}{p^+(x-z)^2+k^+(x-w)^2}$ and the phase $e^{-ik\cdot(y'-z)p^+/k^+}$. Then, exactly,

$$\int_\Lambda^{\vee-k^+}\frac{dp^+}{p^+}\,g(p^+)=\int_\Lambda^{\vee-k^+}dp^+\,\frac{g(p^+)-g(0)}{p^+}+g(0)\Big[\log\frac{\vee}{\Lambda}+\log\Big(1-\frac{k^+}{\vee}\Big)\Big]. \tag{4.1}$$

Your log terms agree with this exactly; the coefficients and the $p^+=0$ phase $e^{-ik\cdot(y'-w)}$ are right. Your remaining integrals, however, run over $0<p^+<\vee-k^+$, both for the subtracted $\frac{1}{p^+}$ term and for the regular terms ($\frac{\delta\delta}{p^++k^+}$, $\frac{\delta\delta}{k^+}$, and so on). The difference is $-\int_0^\Lambda dp^+(\dots)$. It vanishes only for $\Lambda\to 0$, which is a limit you do not want to take.

- Numerically it is 0.3–10% of the row ($\Lambda/k^+=0.065$, $|\mathbf{k}||y'-z|/k^+\sim 1$).
- With the lower limit Λ the prescription reproduces the exact integral to 10^{-15}.

Fix: $\int_0^{\vee-k^+}\to\int_\Lambda^{\vee-k^+}$ in Eqs. III(1.7), III(1.10), III(1.13), III(1.14), III(1.17), with $[\frac{1}{p^+}]_+$ defined on $(\Lambda,\vee-k^+)$ as above.

5 M3: Group II, Row IV

Lines II(1.12)–II(1.14) are correct: the explicit form, the w and w' integration with the Jacobian $(\frac{p^++k^+}{k^+})^4$, the polarization sum, the phase, and the colour reduction to N_c.

The z integral. The remaining z dependence is the tensor integral $J^{mm'}=\int_z\frac{(y-z)^m(y'-z)^{m'}}{(y-z)^2(y'-z)^2}$. It is UV finite and IR (collinear) divergent. In $d_\perp=2-\epsilon$, with $(\mu^2e^{\gamma_E}/4\pi)^{\epsilon/2}$ (the scheme of Group I Row II), the Fourier representation gives

$$J^{mm'}=-\frac{\pi}{2}\,\delta^{mm'}\Big[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2}{4}\Big]-\pi\,\frac{(y'-y)^m(y'-y)^{m'}}{(y'-y)^2}+O(\epsilon). \tag{5.1}$$

The trace in $d_\perp$ dimensions is $-\pi[\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2}{4}]$.

- The traceless part was checked numerically in two dimensions.
- $J^{mm'}=J^{m'm}$, so the term $\frac{2}{k^+}(\delta_{mk'}\delta_{m'k}-\delta_{m'k'}\delta_{km})$ of II(1.14) drops out.
- The $d_\perp$ of the first term, times the pole, leaves a finite -1.

The result, with $\xi=\frac{k^+}{p^++k^+}$ and the exact limits $\frac{k^+}{\vee}<\xi<\frac{k^+}{k^++\Lambda}$, is

$$\begin{aligned}\frac{d^3N}{d^3k}=&-\frac{1}{(2\pi)^3}\,\frac{g^4N_c}{8\pi^4}\,\frac{1}{k^+}\int_{k^+/\vee}^{k^+/(k^++\Lambda)}\frac{d\xi}{\xi^2}\int_{x,x',y,y'}e^{-ik\cdot(y'-y)/\xi}\Bigg\{\frac{(x-y)\cdot(x'-y')}{(x-y)^2(x'-y')^2}\Bigg[\Big(\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big)\\&\times\Big(\frac{2}{\epsilon}+2\gamma_E+\log\frac{(y'-y)^2\mu^2}{4}\Big)-\xi(1-\xi)\Bigg]+2\xi(1-\xi)\,\frac{(x-y)\cdot(y'-y)\,(x'-y')\cdot(y'-y)}{(x-y)^2(x'-y')^2(y'-y)^2}\Bigg\}\\&\times\Big[U^{ce'}(y')-U^{ce'}(x')\Big]\rho^{e'}(x')\Big[U^{ce}(y)-U^{ce}(x)\Big]\rho^e(x).\end{aligned} \tag{5.2}$$

This is algebraically the same as your second and third terms (your "+1" and your third term combine into $-\xi(1-\xi)$ and the tensor term), apart from the three points below.

(a) The constant in the logarithm. In your sign convention ($\frac{1}{\bar\epsilon}$ for the infrared pole) the bracket is $-\big[\frac{1}{\bar\epsilon}-\log\frac{(y'-y)^2\mu^2e^{2\gamma_E}}{4}\big]$, not $\big[\frac{1}{\bar\epsilon}-\log(4\pi^2\mu^2(y'-y)^2)\big]$. The scale must be the same $\mu_{\overline{MS}}$ as in Group I Row II, so that the poles of the different rows can be combined.

(b) The endpoint. With $\frac{\xi}{1-\xi}=\frac{1}{1-\xi}-1$ and the exact identity $\int_{\xi_0}^{\xi_1}\frac{g(\xi)}{1-\xi}=\int_{\xi_0}^{\xi_1}\frac{g(\xi)-g(1)}{1-\xi}+g(1)\log\frac{1-\xi_0}{1-\xi_1}$, the soft endpoint carries

$$\log\frac{1-\xi_0}{1-\xi_1}=\log\frac{(\vee-k^+)(k^++\Lambda)}{\vee\Lambda}=\log\frac{\vee}{\Lambda}-\log\frac{\vee}{k^+}+\log\Big(1-\frac{k^+}{\vee}\Big)+\log\Big(1+\frac{\Lambda}{k^+}\Big). \tag{5.3}$$

Your first term has only the first two logarithms, and your $\int_1^{k^+/\vee}$ has $\xi_1=1$ in place of $\frac{k^+}{k^++\Lambda}$.

(c) The fragmentation form. Let $d^3N_{\rm LO}(k^+,\mathbf{k})$ be the LO cross section with emitter y. The pole part of this row is then

$$\left.\frac{d^3N}{d^3k}\right|_{\rm pole} = -\frac{\alpha_s}{2\pi}\,\frac{2}{\epsilon}\int_{k^+/\vee}^{k^+/(k^+\!+\Lambda)}\frac{d\xi}{\xi^3}\,P_{gg}^{(r)}(\xi)\,\frac{d^3N_{\rm LO}}{d^3k}\Big(\frac{k^+}{\xi},\frac{\mathbf{k}}{\xi}\Big), \qquad P_{gg}^{(r)} = 2N_c\Big[\xi(1-\xi)+\frac{\xi}{1-\xi}+\frac{1-\xi}{\xi}\Big]. \tag{5.4}$$

- The measure is $d\xi/\xi^3$: the LO cross section at k^+/ξ carries ξ/k^+.
- This row supplies only the real part ($\xi<1$) of the collinear counterterm. The $\delta(1-\xi)$ part of P_{gg} ($\beta_0/2$ and the soft endpoint) comes with the virtual terms.
- Your II(1.16), with $D_{g/g}=\delta(1-\xi)-\frac{g^2}{8\pi^2}\frac{1}{\epsilon}P_{gg}$ under the integral, therefore adds $\pm$LO (the δ term, at the end point of the integration) to a row that does not contain it.

The corrected derivation is in `Group_II_corrected.tex`.

6 M4: index placement for a general adjoint U

These placements make every line agree with the first-principles evaluation for a general (non-symmetric) adjoint U (Sec. 9). With your convention $U^{bc}=U^{cb}$ the written and the corrected forms are equal.

where	as written	for a general U
one-ρ $\mathfrak{B}$, amplitude: II-I(1), II-I(2), III-I, III-II, III-III(2)	$f^{acd}U^{bc}(z)U^{de}(x)\rho^{e}(x)$	$f^{acd}U^{cb}(z)U^{de}(x)\rho^{e}(x)$
one-ρ $\mathfrak{B}$, conjugate: II-I(1) block B	$f^{ac'd'}U^{bc'}(z)U^{d'e'}(x)\rho^{e'}(x)$	$f^{ac'd'}U^{c'b}(z)U^{d'e'}(x)\rho^{e'}(x)$
one-ρ $\mathfrak{B}$, conjugate: II-I(2)	$f^{ac'd'}U^{bc'}(z)U^{e'd'}(x')\rho^{e'}(x')$	$f^{ac'd'}U^{c'b}(z)U^{d'e'}(x')\rho^{e'}(x')$
II-I(2), N_c term II(1.4), II(1.5)	$N_cU^{e'd}(x')U^{de}(x)\rho^{e'}\rho^{e}$	$N_cU^{de'}(x')U^{de}(x)\rho^{e'}\rho^{e}$
two-ρ $\mathfrak{B}$: II-I(1), III-III(1)	$U^{bc'}(z)U^{ad'}(x')U^{c'e'}(y)$	$U^{c'b}(z)U^{ad'}(x')U^{c'e'}(y)$
$\mathbb{X}_1'$, conjugate: III-I	$U^{bd'}(z)U^{d'e'}(y)U^{ac'}(w')$	$U^{d'b}(z)U^{d'e'}(y)U^{ac'}(w')$
$\mathbb{X}_2'$, conjugate: III-II	$U^{c'd'}(y)U^{ab'}(x')U^{bc'}(z)$	$U^{c'd'}(y)U^{ab'}(x')U^{c'b}(z)$
$\mathbb{X}_1'$, amplitude: III-V	$U^{bc}(z)U^{ce}(x)U^{ad}(y)$	$U^{cb}(z)U^{ce}(x)U^{ad}(y)$
$\mathbb{X}_2'$, amplitude: III-VI	$U^{bc}(z)U^{ad}(y)U^{ce}(x)$	$U^{cb}(z)U^{ad}(y)U^{ce}(x)$
$\mathfrak{C}'$, conjugate: III-III, III-V, III-VI, II-IV	$U^{e'c'}(y')U^{d'b}(z)\rho^{e'}(x') - U^{bd'}(z)U^{c'e'}(x')\rho^{e'}(x')$	$U^{c'e'}(y')U^{d'b}(z)\rho^{e'}(x') - U^{d'b}(z)U^{c'e'}(x')\rho^{e'}(x')$
$\mathfrak{C}'$, amplitude: II-IV	$f^{cda}U^{ec}(y)U^{db}(z)\rho^{e}-f^{cda}U^{bd}(z)U^{ce}(x)\rho^{e}$	$f^{cda}U^{ce}(y)U^{db}(z)\rho^{e}-f^{cda}U^{db}(z)U^{ce}(x)\rho^{e}$
II-IV after the reduction II(1.14)–II(1.16)	$[U^{e'c}(y')-U^{e'c}(x')][U^{ec}(y)-U^{ce}(x)]\rho^{e'}\rho^{e}$	$[U^{ce'}(y')-U^{ce'}(x')]\rho^{e'}(x')[U^{ce}(y)-U^{ce}(x)]\rho^{e}(x)$
III-V after the simplification III(1.14)	$U^{e'c'}(y')\rho^{e'}(x')$	$U^{c'e'}(y')\rho^{e'}(x')$

The simplifications of Row V III(1.14) and of Row IV II(1.14) use $\sum_b U^{d'b}(z)U^{cb}(z)=\delta^{d'c}$. This holds for a general U only with the placements in the right column. With the second term of $\mathbb{X}_1$ as written in `Calculation_main.tex` it would give $(U^2)^{d'c}$; see M6.

7 M5: integration measures

- II(1.3), II(1.4), II(1.5): $\int_{z,w,w'}\to\int_{x,x',z,w,w'}$.
- II(1.12): $\int_{x,y,y'}\to\int_{x,x',y,y'}$.
- II(1.13), II(1.14): $\int_{z,w,w'}\to\int_{z}$ (w,w' are integrated by the δ-functions), and $\int_{x,y,y'}\to\int_{x,x',y,y'}$. In II(1.13) $k\{+\}\to k^+$.
- II(1.15), II(1.16): $\int_{w,w'}$ is no longer an integration; the variables are x,x',y,y'.
- III(1.1), III(1.3): $\int_{z,w,w'}\to\int_{x,y,x',z,w,w'}$.
- III(1.9), III(1.10): $\int_{w,w',y',z,y,x}\to\int_{w,x',y',z,y,x}$ (w' is integrated, x' is missing).
- III(1.10) log term: $\int_{w,y',z,y,x}\to\int_{w,x',y',z,y,x}$.
- III(1.15): x' is missing.

8 M6: points in `Calculation_main.tex`

1. $\mathbb{X}_1$, **second term.** $-U^{ac}(w')\bar{\mathbb{A}}^d_j(q^+,-v)U^{\dagger db}(v)\mathbb{A}^c_i(k^+,-w')$ should have $U^{db}(v)$.
   - The unobserved gluon is created after the target by $\bar{a}^{\dagger d} = U^{de}a^{\dagger e}$, and $\langle g^b|a^{\dagger e}|0\rangle \propto \delta^{eb}$, so the factor is $U^{db}(v) = U^{\dagger bd}(v)$.
   - This agrees with the first term of $\mathbb{X}_1$, with $\mathbb{X}_2$, $\mathfrak{B}$ and $\mathfrak{C}$, which all have $U^{\dagger b\cdot}(v)$.
   - Only with this placement do the $U(z)$ cancel in Row V. Irrelevant if $U = U^T$.
2. **Range of $\mathbb{B}_2$.** $\mathbb{B}^{ba}_2(k^+,k;p^+,p)$ is written with $\Theta(k^+-\Lambda-p^+)$. This is the ξ-form $\Theta(1-\Lambda/k^+-\xi)$ with k^+ the total momentum, carried over to the form in which k^+ and p^+ are the two gluon momenta. There it should read $\Lambda < p^+$, $\Lambda < k^+$, $p^+ + k^+ < \vee$. Your Group II and III notes use the correct range $\Lambda < p^+ < \vee - k^+$.
3. **(CS2), $\mathbb{X}_2$ in the conjugate amplitude.** $\bar{\mathbb{A}}^\dagger\bar{\mathbb{A}}^\dagger U^{bc'}(z)$ should be $U^{c'b}(z)$. The same slip appears in your Group III Row II (Sec. 6).
4. **(A1coeff).** The argument should be $-\mathbb{A}^a_{2i}(k^+,-\mathbf{z}) = \mathbb{A}^{\dagger a}_{1i}(k^+,\mathbf{z})$, as in the unitarity relation $\mathbb{A}^{(1)\dagger}_1(p^+,z) = -\mathbb{A}^{(1)}_2(p^+,-z)$. The formulas used afterwards are consistent with this.

9 Checks

All numbers come from `checks23.py` (`checks23.json`). The code is in `fp.py` (first principles), `user23*.py` (your equations as written) and `fixed23.py` (the placements of Sec. 6). Parameters: $k^+ = 1.7$, $\Lambda = 0.11$, $\vee = 23$, $g = 1.3$, SU(3).

check	result
Before the integrations: written / first principles (3 random configurations each)	
II-I part 1, block A II(1.1)	$U = U^T$: 1; general U: 2.91, -0.05, 0.44; with Sec. 6 placements: 1 ± 0
II-I part 1, block B II(1.1)	$U = U^T$: 1; general U: 6.62, 0.26, -2.11; with Sec. 6 placements: 1 ± 0
II-I part 2 II(1.3)	$U = U^T$: 1; general U: -3.79, 0.26, 1.07; with Sec. 6 placements: 1 ± 0
III-I III(1.1)	$U = U^T$: 1; general U: -1.53, 1.18, -3.39; with Sec. 6 placements: 1 ± 0
III-II III(1.3)	$U = U^T$: 1; general U: -1.29, 0.77, 0.27; with Sec. 6 placements: 1 ± 0
III-III part 1 III(1.5)	$U = U^T$: -1; general U: -0.09, 1.73, 0.07; with Sec. 6 placements: 1 ± 0
III-III part 2 III(1.8)	$U = U^T$: -1; general U: -0.45, -0.71, -0.93; with Sec. 6 placements: 1 ± 0
III-V III(1.11)	$U = U^T$: -1; general U: -1.67, 2.06, 0.04; with Sec. 6 placements: 1 ± 0
III-VI III(1.15)	$U = U^T$: -1; general U: 9.65, 3.02, -2.51; with Sec. 6 placements: 1 ± 0
II-IV II(1.12)	$U = U^T$: 1; general U: -0.01, 3.43, 2.03; with Sec. 6 placements: 1 ± 0
Integrated lines: —written/exact -1— (general U, Sec. 6 placements)	
II-I part 1, p^+ integrated II(1.2)	4.4×10^{-16}
II-I part 2, 2a	2.2×10^{-16}
II-I part 2, 2b	2.8×10^{-15}
II-I part 2, 2c	6.7×10^{-16}
II-I part 2, 2d	9.7×10^{-14}
II-I part 2, 2e	2.8×10^{-13}
II-I part 2, 2f	6.5×10^{-11}
II-I part 2, kernel of II(1.5) / kernel of II(1.3)	4.4×10^{-16}
II-I part 2, colour of II(1.4) / colour of II(1.3)	1.0×10^{-14}
III-I, p^+ integrated III(1.2); III(1.4) has the same kernel	4.4×10^{-16}
$\mathfrak{C}$ rows of Group III: —written/exact -1— (min–max over 3 configurations)	
w' integration, III-III(1) III(1.6)	0 – 6.7×10^{-16}
w' integration, III-III(2) III(1.9)	0 – 1.1×10^{-15}
w' integration, III-V III(1.12)	0 – 6.7×10^{-16}
w' integration, III-VI III(1.16)	0 – 6.7×10^{-16}
plus prescription, III-III(1) III(1.7): as written ($\int_0$)	3.4×10^{-3} – 8.0×10^{-2}

plus prescription, III-III(1) III(1.7): lower limit Λ	0 – 2.5×10^{-16}
plus prescription, III-V III(1.13): as written ($\int_0$)	3.4×10^{-3} – 8.0×10^{-2}
plus prescription, III-V III(1.13): lower limit Λ	0 – 2.5×10^{-16}
plus prescription, III-VI III(1.17): as written ($\int_0$)	3.4×10^{-3} – 8.0×10^{-2}
plus prescription, III-VI III(1.17): lower limit Λ	0 – 2.5×10^{-16}
plus prescription, III-III(2) III(1.10): as written ($\int_0$)	1.4×10^{-2} – 1.0×10^{-1}
plus prescription, III-III(2) III(1.10): lower limit Λ	0 – 1.2×10^{-15}
Row V simplification and Group II, Row IV	
III-V: colour of III(1.14) / III(1.13), |ratio − 1|	3.2×10^{-14}
II-IV: II(1.13) / [II(1.12) at w^*, w'^* $\times(\frac{p^+ + k^+}{k^+})^4$], |ratio − 1|	1.4×10^{-15}
II-IV: colour of II(1.14) / II(1.13), |ratio − 1|	6.7×10^{-16}
$J^{mm'}$: traceless part in 2 dims vs. $-\pi(\hat{s}^m \hat{s}^{m'} - \frac{1}{2}\delta^{mm'})$; symmetry $\|J^{12} - J^{21}\|$	1.1×10^{-15}; 1.3×10^{-15}
II-IV: polarization sum $\times J^{mm'}$ ($d_\perp = 2 - \epsilon$) vs. the corrected bracket (Sec. 5), relative	3.2×10^{-12}
$k^+(\frac{p^+}{(p^+ + k^+)^2} + \frac{p^+}{k^{+2}} + \frac{1}{p^+}) - \big[\xi(1-\xi) + \frac{\xi}{1-\xi} + \frac{1-\xi}{\xi}\big]$; endpoint logarithm; LO normalization of the pole (`sympy`)	0; 0; 0